

Baselines 10718

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[Github link](#)

1 Part 1: Revised Proposal

Note: We rewrote the majority of our proposal, including points that did not receive a 1, in order to have a better structure off which to build our system.

1.1 Problem

Original:

We are solving the problem of reviewing machine learning papers so researchers in the field can understand shortcomings of their papers and what to improve to increase the likelihood of their paper being accepted. Currently many people use human feedback which could be biased. This is targeted towards all ML researchers and affects universities and other organisations with people trying to publish ML papers.

Researchers submitting to ML conferences receive formal feedback during review, with a short rebuttal window in which to respond and revise. That window rarely leaves time for substantial changes such as new baselines, additional experiments, or reworked analyses, so the weaknesses that most affect acceptance are often the hardest to fix once reviews arrive. The feedback is also noisy: in the NeurIPS 2021 consistency experiment, two independent committees disagreed on 23% of accept/reject decisions, and about half of the accepted papers would have changed had the process been rerun (Beygelzimer et al., 2023).

Before submission, the main alternative is feedback from mentors and peers. This costs those mentors time, is unavailable to many early-career researchers and those outside well-resourced labs, and tends to miss issues an outside reviewer would raise, because colleagues share the author's context and assumptions. We aim to give researchers reviewer-style feedback on a draft before they submit, when there is still time to act on substantial changes.

We added more details to the scenario when this should be used

1.2 Goal

Original:

We are trying to change the likelihood of getting realistic and actionable insight out of an AI reviewer.

Realistic: This may be measured by comparing statistics on scores and acceptance decisions assigned by humans and our system.

Actionable: We may inject analytic and conceptual mistakes of varying levels of severity into an accepted paper and estimate the likelihood of our system catching them.

We are trying to change the number of substantive weaknesses (such as missing baselines, insufficient experiments, or unsupported claims) and basic oversights (such as formatting problems or missing details) that formal reviewers raise about a submitted paper. These weaknesses are realistic problems that the user can act on, by addressing each weakness with edits. The changes are measured by the count of these issues in the official reviews of papers whose authors used our system before submission, compared with similar papers whose authors did not.

We address the feedback we received by adding details about the system's outputs being realistic and actionable.

1.3 Action

Original:

The researcher will read that feedback and run experiments and/or make edits to their paper. This can happen multiple times in an interactive process.

Several weeks before a submission deadline, the researcher runs the system on their draft and receives a ranked list of up to five issues and a severity rating. For each issue, the researcher decides whether to fix it, acknowledge it in the paper (for example, in the limitations section), or dismiss it as incorrect. They schedule substantial fixes, such as new baselines or experiments, first, since these cannot be done during a rebuttal period, and handle quick fixes such as missing details or formatting alongside them. After revising, they re-run the system on the new draft to confirm the issues are resolved and to surface the next set, repeating weekly until no high-severity issues remain or the deadline arrives.

We improved this by adding more details about realistic scenarios that users would be facing, and how the tool is used.

1.4 Capacity

Original: Capacity:

The researcher can act on 5 potential improvements to the paper per week.

Justification:

Substantial improvements to papers take overhead to review. It can take a week to respond to typical conference reviews. If our system surfaces real improvements changes must be made as well as reviewed additionally.

1.4.1 Number

The researcher can act on up to five substantial issues per week, such as adding new baselines or experiments.

1.4.2 Justification

Five issues roughly matches the number of weaknesses in a typical formal review, so each weekly run gives about one review's worth of feedback. Substantial fixes such as new experiments take several days of implementation, compute, and writing, so 5 per week is realistic alongside a researcher's other work. Longer lists dilute attention and make it harder to tell the critical issues from the minor ones, so we rank issues by severity and show only the top five. Over a four-to-six-week pre-submission window, this allows 20–30 issues to be addressed, more than the distinct issues reviewers raise on a typical paper in our data.

We add more details about our specific capacity and reasoning about why this is a suitable capacity, such as how the system is intended to be used.

1.5 Data

Original:

We will use public reviews of real machine learning papers, specifically the PeerRead dataset.

We have this dataset today.

We use conference submission forums from OpenReview. There exist various pre-processed corpora such as PeerRead, CASIMIR, ACL [3, 2, 1], which we augment with OpenReview forums from ICLR that are not already covered in the datasets we consider. This includes text transcriptions of papers, as well as full discussion comment chains between reviewers and authors. The usage of the data is further elaborated in later sections.

We incorporate feedback from Namrata such as including the CASIMIR dataset

1.6 ML Analysis Type

Original: Describe, Predict, Generate

Our system performs **prediction** and **generation**. Given a paper draft, it predicts the weaknesses that formal reviewers at the target venue would raise, learned from official reviews of comparable papers, and ranks them by predicted severity. For each of the top five, it generates a description of the issue tied to a specific location in the paper (section, table, or claim). The system does not edit the draft itself: the researcher decides whether to fix, acknowledge, or dismiss each issue (see Action).

We add a lot more refinement to our initial response on what we meant by describe, predict and generate.

1.7 How would we know it worked?

Original:

We can measure performance of the model by comparing model outputs and real reviews that a researcher receives after submitting a paper. 3 months after launch, we would have initial comparisons, after six months we will have rebuttals, and after a year, we would have many papers with research.

Our goal, fewer weaknesses raised in formal reviews, can only be measured after deployment. Within the semester, we test its prerequisite: whether the system, run on a draft, flags the weaknesses real reviewers go on to raise. Our ground truth is official conference reviews, and the decisions made in these reviews along with the feedback.

Offline evaluation. We report P@5 for our system and both baselines on the held-out ICLR 2026 set.

Prospective evaluation. When ICLR 2027 submissions and reviews are released in November, we will run all systems on a sample of these papers and compute P@5 against their reviews. Because these reviews postdate the training cutoff of every model we use, this rules out inflated results from LLMs that memorized older public reviews, and provides a clean test set.

We address the feedback we got about the evaluation pipeline by adding information about who/what source the papers will be from, and how they will be used for evaluation of the pipeline in a timely matter.

2 Evaluation Setup

2.1 Data Splits

Across the venues covered in our global corpus, ICLR is the only one that mandatorily requires all submissions’ forums be publicly accessible. For other venues, the typical protocol is to release forums for accepted papers and allow rejected papers’ authors to opt-in. To avoid training on an unrealistically optimistic corpus, we rely heavily on the ICLR dataset.

We hold-out one ICLR cycle, 2026 as a test set. We aim to verify our method is robust to the distribution shift amongst reviewers and papers in the next year. This will offer insight into model performance in the coming cycle. We also plan to use 2025 as a validation set, with all other years as potential training data.

We also plan to evaluate against CASIMIR [2], which better models an iterative review process but leave this out of scope for the baseline. This dataset is intended to be used as a sanity check, and will not be used directly for model selection.

2.2 Metric and Constraints

Metric of choice. We choose our metric to be the precision in the top 5 highest severity issues. Since the researcher’s capacity is 5 per week, this metric will highlight how much of our recommendations that the researcher will work on would actually matter to a real reviewer. Formally:

For a paper p , let R_p be the deduplicated weaknesses raised in its official reviews, and let $S_p = \text{top}_5(M_p)$ be the five highest-severity issues in the system’s predicted issues M_p . We match issues in S_p to weaknesses in R_p one-to-one, so a single generic issue cannot claim several reviewer weaknesses: among pairs (s, r) with $\text{sim}(s, r) \geq \tau$, we choose the largest set of pairs in which each issue and each weakness appears at most once. Here sim is the cosine similarity of sentence embeddings (all-MiniLM-L6-v2) and $\tau = 0.5$. Let $T_p \subseteq S_p$ be the matched issues. Then

$$P_5(p) = \frac{|T_p|}{|S_p|} = \frac{|T_p|}{5},$$

and we report the mean of $P_5(p)$ over test papers. When severity ties make $\text{top}_5(M_p)$ ambiguous, we report the expected value of $P_5(p)$ over all equally likely ways of breaking the ties.

Matching. For first-pass results we match a generated issue to a reviewer weakness when the cosine similarity of their sentence embeddings (all-MiniLM-L6-v2) is at least 0.5, chosen by manually inspecting matched pairs. We will replace this with an LLM judge, validated against human labels, for final results (compute constraints on implementing this currently).

3 Baselines

3.1 Non-ML Baseline

Original: The simplest practice that partly solves this would be a checklist for important components needed in a research paper for submission. ML adds specificity to the paper and provides potential changes that go beyond just a generalized checklist of items to include.

Our non-ML baseline tests whether reviewers raise the same issues on similar papers. For each test paper, we retrieve the ten most similar ICLR 2024 papers by TF-IDF cosine similarity of their titles and abstracts. We extract weakness sentences from these neighbors’ reviews with hand-written rules (headers such as “Weaknesses”, cue words such as “lacks”) and sort them into ten fixed categories (baselines, experiments, clarity, novelty, soundness, analysis, related work, reproducibility, motivation, efficiency) using keyword lists. Each category gets a severity from how harshly the reviewers who raised it scored their papers, $\text{clip}(\text{round}(8 - \bar{r}), 1, 5)$, where $\bar{r}$ is their similarity-weighted mean rating. We output the five most severe categories, breaking ties at random and reporting the expected score over all tie-breaks. Each issue is the reviewer sentence from that category that best matches the test paper (highest TF-IDF similarity to its title and abstract), so the output is real reviewer feedback from the most similar papers.

We replaced the generic checklist, which gives every paper the same feedback, with retrieval of reviewer feedback on similar past papers.

3.2 LLM Baseline

Original:

A simple ML baseline could be to use a 0 shot LLM to evaluate a paper. In order for our approach to be worth deploying, it would require providing insights that go beyond the surface level responses that would be provided through this method.

We give an off-the-shelf LLM the full paper text with a plain prompt asking for its five most important weaknesses, each with a severity score from 1 to 5, and rank by severity. This is what many researchers already do before submitting, so our system must beat it to justify anything more complex.

We give more reasoning to our LLM baseline and why we picked it, and we give the prompt in the next section.

4 First Pass Results

4.1 Non-ML Baseline

We run the baseline on all 13,702 ICLR 2026 submissions and match against their official reviews. We a precision@5 score of 15.8%.

4.2 LLM baseline

We ran Qwen3-8B [4] (4bit quantized, with reasoning mode disabled and greedy decoding) against text extracted from the PDFs of research papers. We ran this against a subsample of 25 papers due to compute restrictions. Because of GPU memory, the input is truncated to 16k tokens. This kept the full main text, up to the references for our subsample.

The prompt is: *“Below is the full text of a paper submitted to ICLR. Acting as a peer reviewer, list the five most important weaknesses of this paper. For each, give a severity from 1 (minor) to 5 (critical). Return only JSON, with no other text.”*, followed by the paper text.

We achieve a precision at 5 score of 29.2% with our LLM baseline. For comparison, the non-ML baseline achieves a P@5 score of 16.9% on this subset.

5 What this tells us

The LLM baseline (29.2% P@5) roughly doubles retrieval performance, so the zero-shot LLM is the bar to beat. At 29% P@5, about 1.5 of a researcher’s 5 weekly issues correspond to something a reviewer raises, so most of their limited fix budget goes to issues that wouldn’t have come up. Notably, this was done with a smaller LLM with reasoning disabled, so frontier models would

likely achieve higher performance. For deployment to be worthwhile, our system should at least reach roughly 50% P@5 (2.5 of 5 issues), so that the majority of each week’s effort targets real reviewer concerns. While this is the minimum performance required for deployment to be worthwhile, we hope our final model is able to exceed this. Our next steps is to build a more complex system to specialize in reviewing papers.

References

- [1] Acl rolling review data collection (arr-dc).
- [2] Leane Jourdan, Florian Boudin, Nicolas Hernandez, and Richard Dufour. Casimir: A corpus of scientific articles enhanced with multiple author-integrated revisions, 2024.
- [3] Dongyeop Kang, Waleed Ammar, Bhavana Dalvi, Madeleine van Zuylen, Sebastian Kohlmeier, Eduard Hovy, and Roy Schwartz. A dataset of peer reviews (peerread): Collection, insights and nlp applications, 2018.
- [4] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025.